

Project Design Phase

Solution Architecture

Date	01 October 2026
Team ID	SWTID-2026-8542
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Solution Architecture:

ComicCraft uses a three-layer architecture. The frontend collects the user's comic requirements, the FastAPI backend coordinates the workflow, and the AI models generate the comic outline, story, dialogue, and illustrations.

- Frontend: HTML, CSS and Jinja2 collect the story prompt, character name, setting, tone and art style.
- Backend: FastAPI receives form or JSON input and coordinates the AI generation workflow.
- AI Integration: Gemini Flash creates the 5-panel outline, Gemini Pro creates narration and dialogue, and Stable Diffusion generates panel illustrations.
- Output: The panels are organized into a layout and exported as a downloadable PDF.

Solution Architecture Diagram:

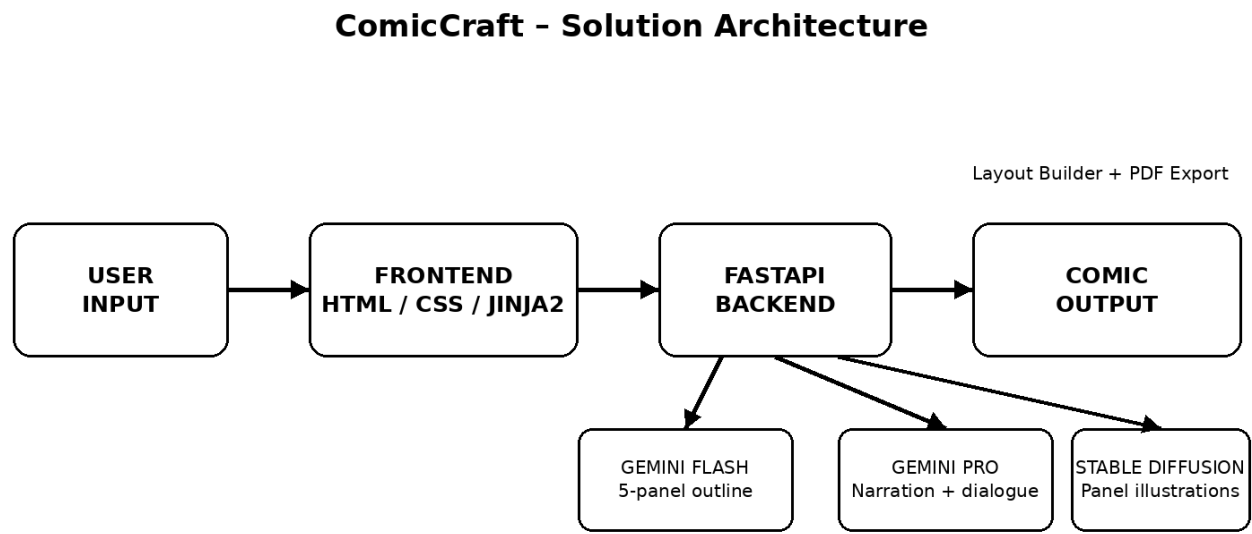


Figure: Architecture and data flow of the ComicCraft AI comic generation application